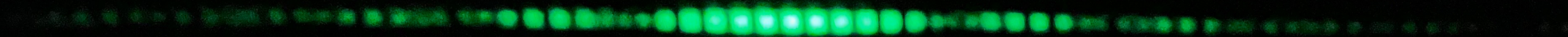
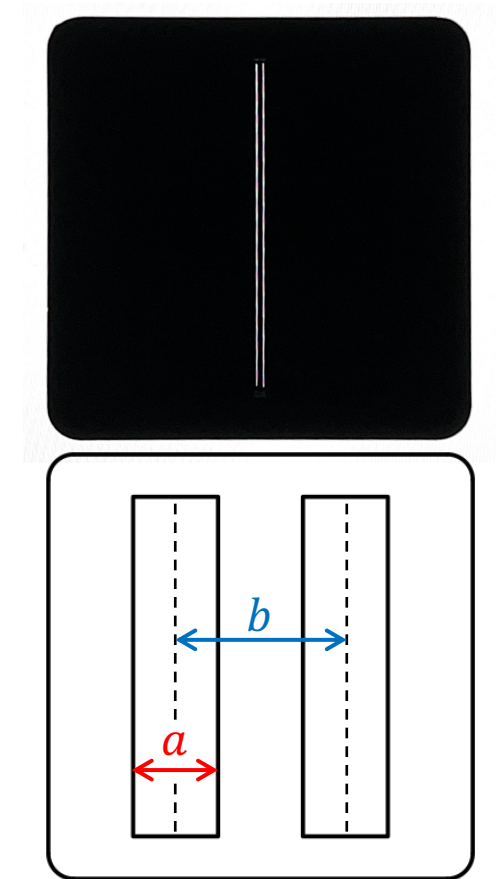
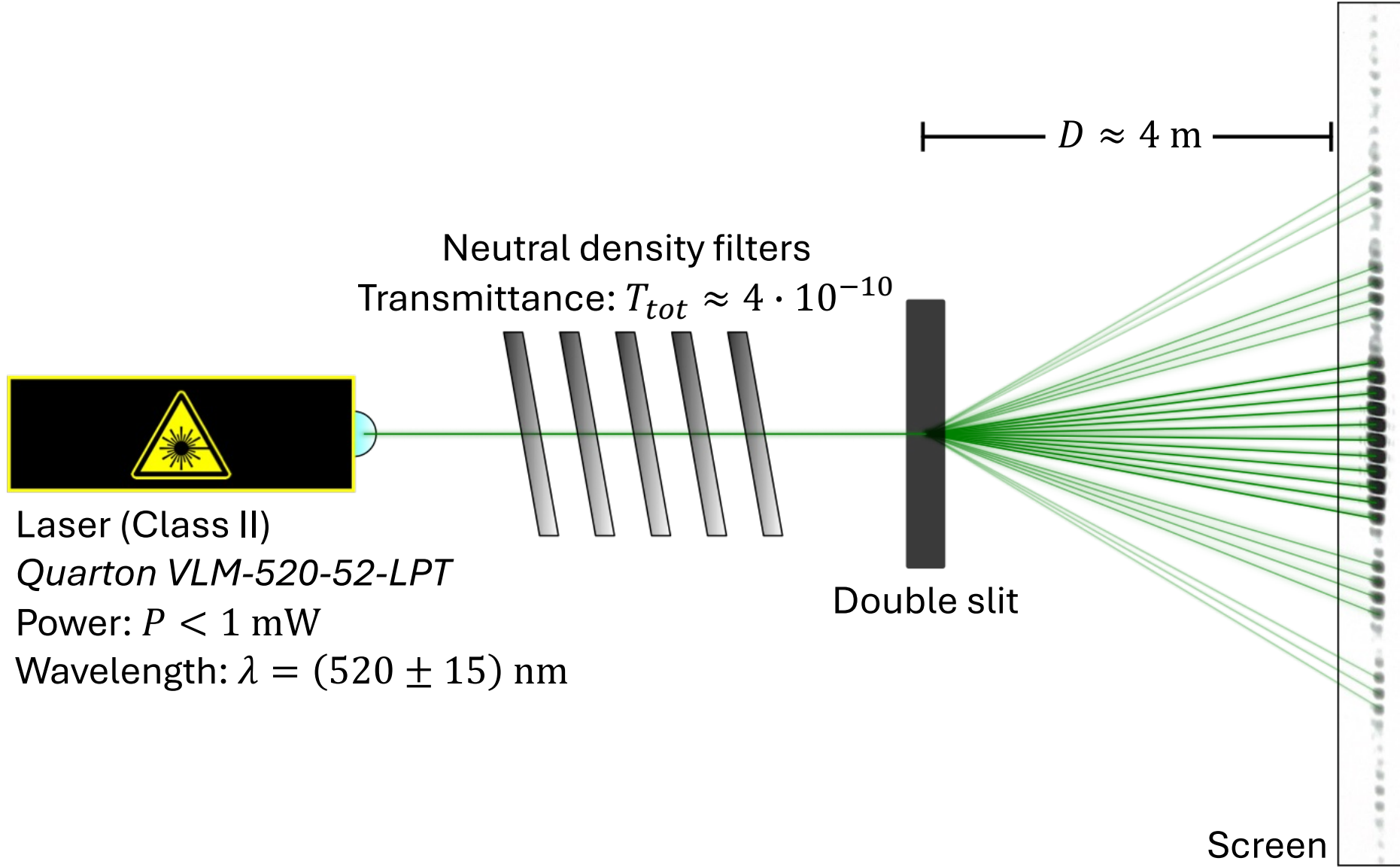


A «single-photon» diffraction experiment built at home using a 3D printer



Experimental setup

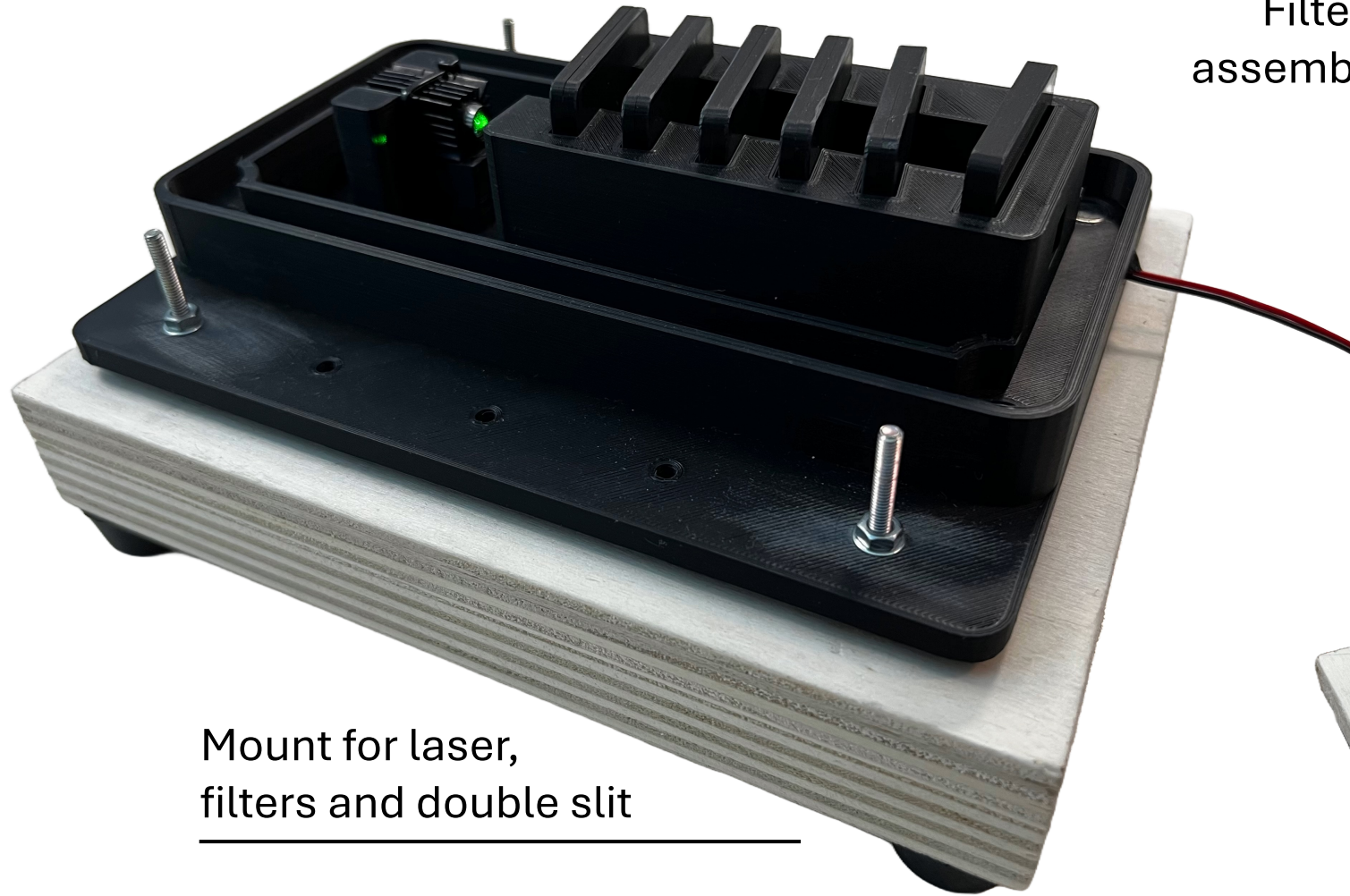


$$a = (0.12 \pm 0.03) \text{ mm}$$

$$b = (0.81 \pm 0.08) \text{ mm}$$

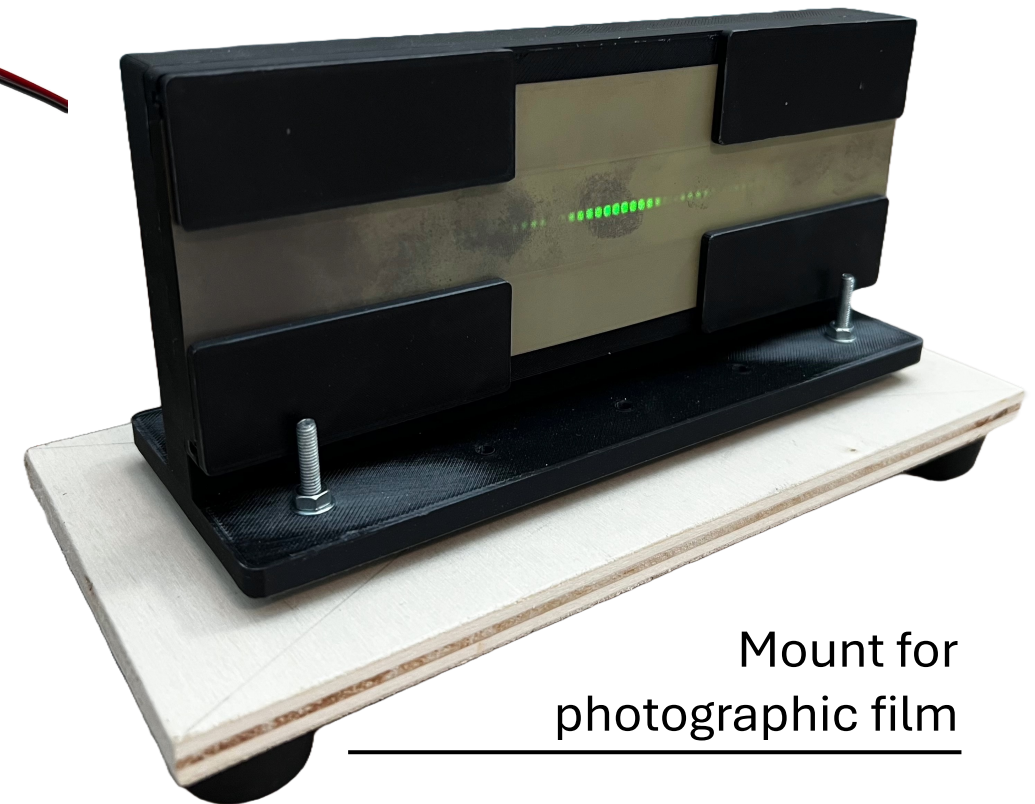
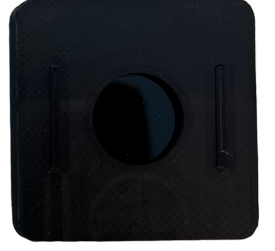
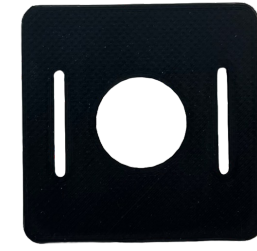
(measured with
a microscope)

Photos of the equipment



Mount for laser,
filters and double slit

Filters
assembly



Mount for
photographic film

«Single-photon» regime

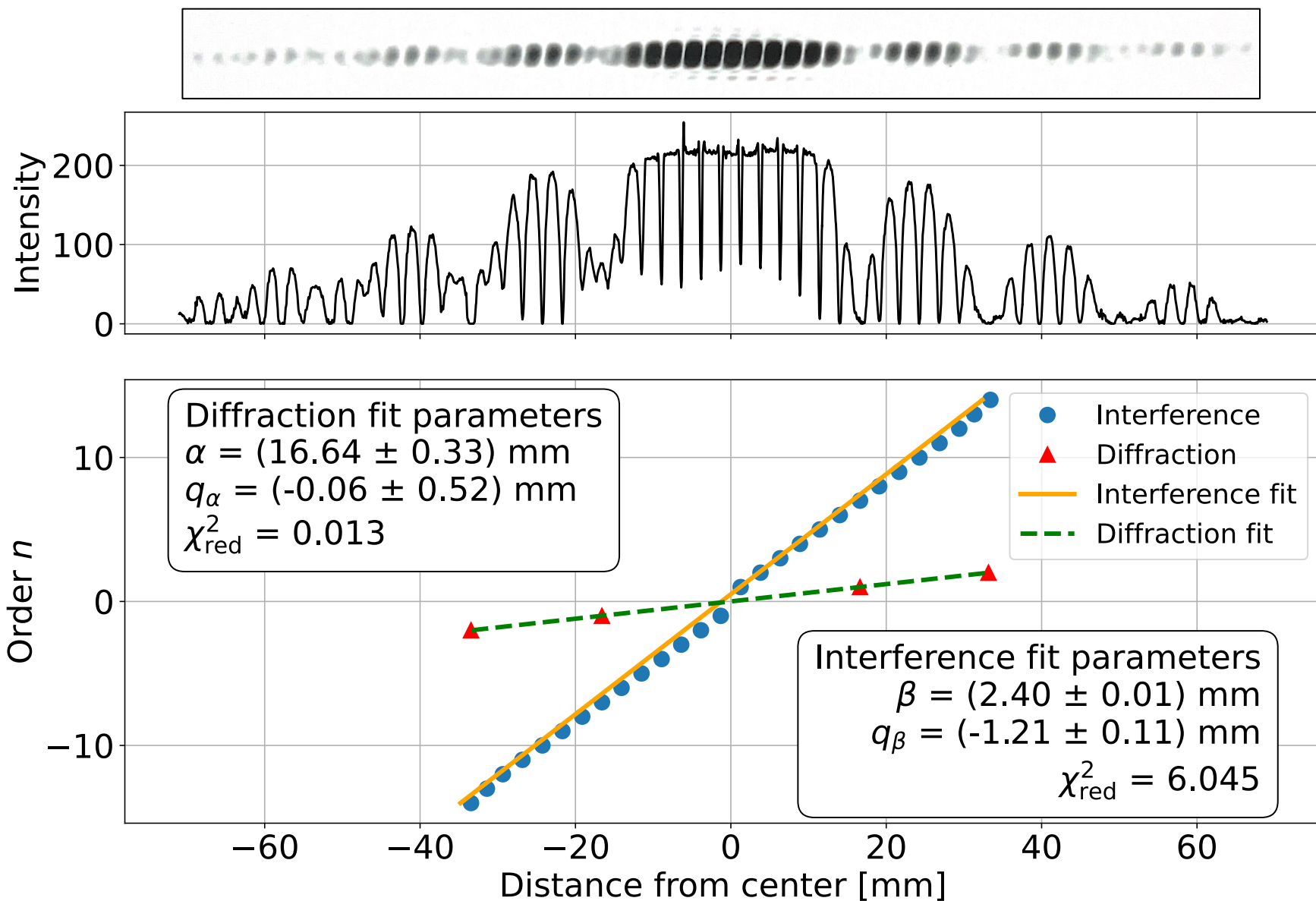
Probability of having n photons per path^[1]: $P(n) = \frac{N^n \cdot e^{-N}}{n!}$

Mean number of photons per path: $N = \frac{E_{laser}}{E_\gamma} \cdot T_{tot} = \frac{P \cdot \Delta t}{h \cdot \nu} \cdot T_{tot} = \frac{P \cdot \frac{D}{c}}{h \cdot \frac{c}{\lambda}} \cdot T_{tot} = \frac{P \cdot D \cdot \lambda}{h \cdot c^2} \cdot T_{tot} < 0.02$

- Laser power: $P < 1$ mW
- Screen distance: $D \approx 4$ m
- Laser wavelength: $\lambda < 535$ nm
- Total transmittance of the filters: $T_{tot} \approx 4 \cdot 10^{-10}$
- Planck constant: $h \approx 6.6 \cdot 10^{-34}$ J · s
- Speed of light in air: $c \approx 2.99 \cdot 10^8$ m/s

Fraction of single photons among all emitted photons: $S(N) = \frac{P(n=1)}{P(n \geq 1)} = \frac{P(1)}{1 - P(0)} = \frac{Ne^{-N}}{1 - e^{-N}}$
|
 $S(N < 0.02) > 99\%$

Results in the high-intensity regime



Positions of the minima^[2]

$$x_{diff} = \frac{\lambda D}{a} \cdot n$$

$$x_{int} = \frac{\lambda D}{b} \cdot \left(n + \frac{1}{2}\right)$$

Fit functions

$$x_{diff}(n) = \alpha \cdot n + q_\alpha$$

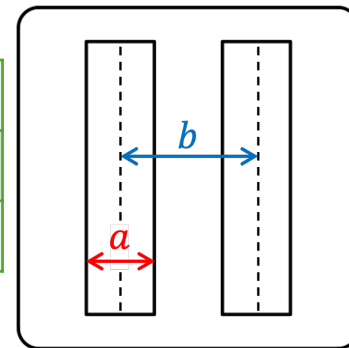
$$x_{int}(n) = \beta \cdot \left(n + \frac{1}{2}\right) + q_\beta$$

Results: $a = (0.13 \pm 0.04) \text{ mm}$
 $b = (0.9 \pm 0.3) \text{ mm}$

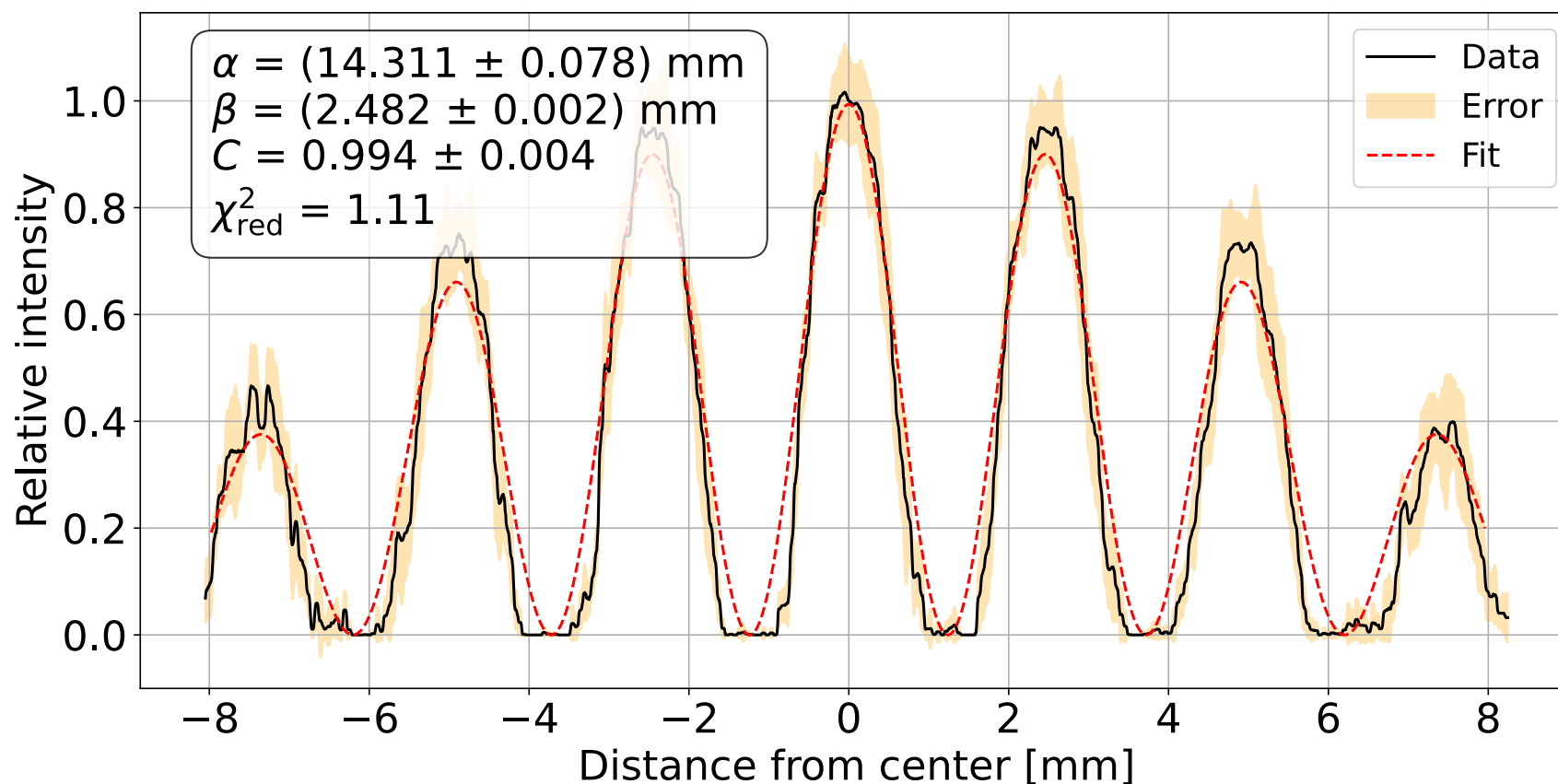
Microscope [mm]

$a = 0.12 \pm 0.03$

$b = 0.81 \pm 0.08$



Results in the «single-photon» regime



Relative intensity^[2]

$$I_{\text{rel}}(x) = \cos^2\left(\frac{\pi b x}{\lambda D}\right) \left[\frac{\sin\left(\frac{\pi a x}{\lambda D}\right)}{\left(\frac{\pi a x}{\lambda D}\right)} \right]^2$$

Fit function

$$f(x) = C \cdot \cos^2\left(\pi \frac{x}{\beta}\right) \left[\frac{\sin\left(\pi \frac{x}{\alpha}\right)}{\left(\pi \frac{x}{\alpha}\right)} \right]^2$$

Results: $a = (0.15 \pm 0.01) \text{ mm}$
 $b = (0.85 \pm 0.03) \text{ mm}$

Microscope [mm]

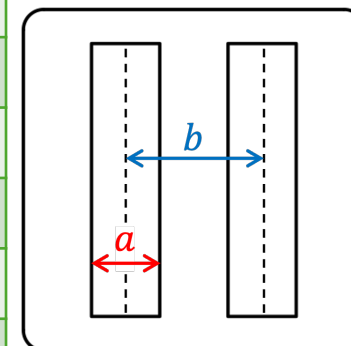
$a = 0.12 \pm 0.03$

$b = 0.81 \pm 0.08$

High-intensity [mm]

$a = 0.13 \pm 0.04$

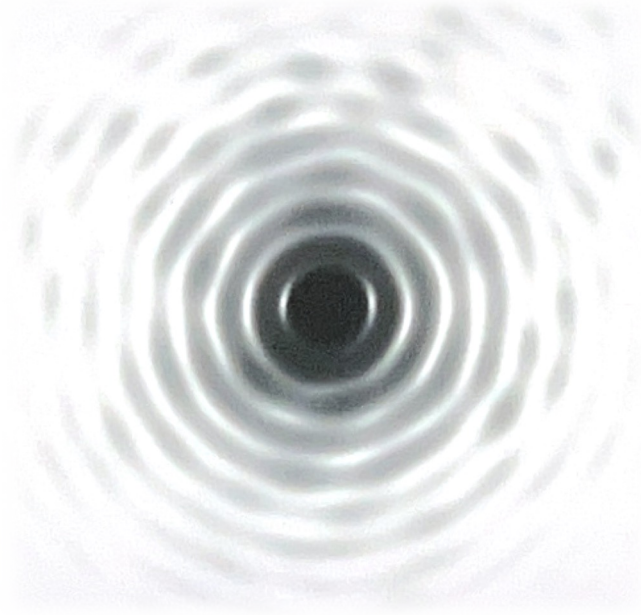
$b = 0.9 \pm 0.3$



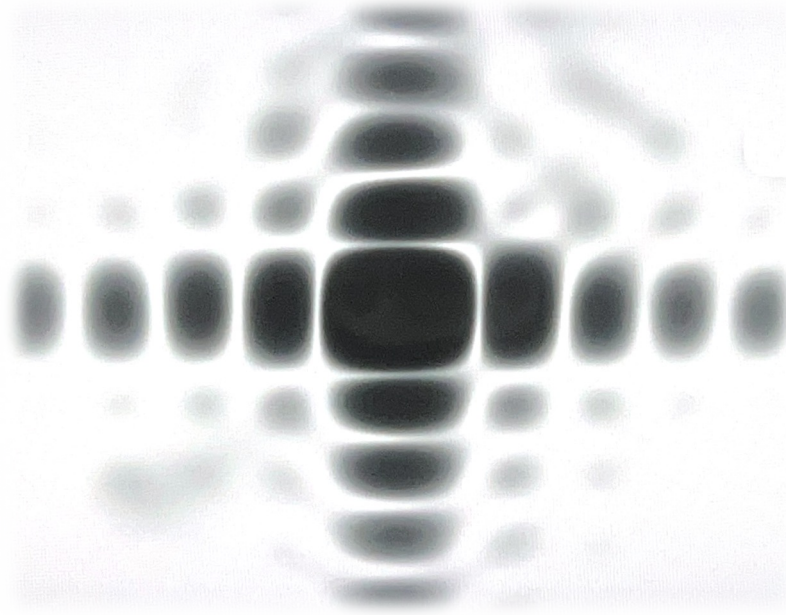
Bonus: further diffraction patterns



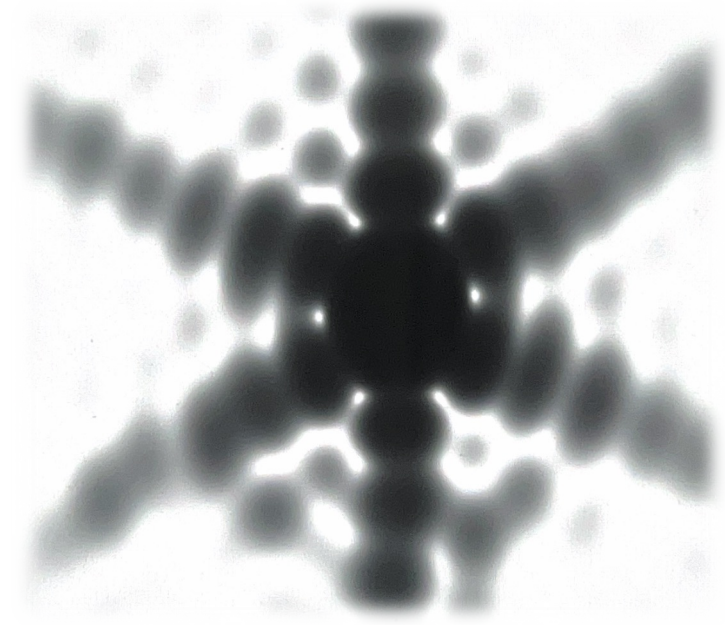
Single slit



Circular pinhole



Square aperture



Triangular aperture

References

- [1] Alan Migdall, Sergey V. Polyakov, Jingyun Fan, and Joshua C. Bienfang, editors. Single-Photon Generation and Detection: Physics and Applications. Academic Press, 2013.
- [2] Eugene Hecht. Optics. Pearson, 5th edition, 2016.